

2024-07-15

SINGLE-PARCEL HEAT DUE DILIGENCE

Site: Diridon Gateway Site (APN-259-27-014)

Lot: 3.34 acres · 140 x 100 m

AOI: 1.20 km² · 329 tiles at 60 m

Parcel peak: 27.31 °C



FORTYGUARD

EXECUTIVE SUMMARY

Diridon Gateway Site (APN-259-27-014) — W Santa Clara St & Cahill St, San Jose CA. 3.34-acre parcel, zoning DC “ Downtown Commercial, proposed use Mixed-use office / residential.

Metric	Value	Reference
Parcel daily peak (area-weighted)	27.31 °C	no city baseline
Parcel daily mean	20.72 °C	AOI mean 20.71 °C
Diurnal swing	10.97 °C	peak minus daily minimum
Hours above 27 °C	35.6 h of 168 h	21% of window
Longest consecutive run	6.5 h	overnight recovery at <12 h
Impervious share (vicinity)	26.9%	EPA trigger >60%
Tree canopy (vicinity)	1.6%	USDA trigger <15%
Heat index at hot hour (13:00)	27.3 °C	NOAA Extreme Caution 32 °C
Peak apparent temperature	24.9 °C	at 13:00
Peak wet-bulb	17.1 °C	evaporative limit 26 °C

Method. A single heatmap was run over the parcel plus a 500 m context ring (1.20 km², 329 tiles at 60 m). Parcel figures are the **area-weighted mean** of the 6 tiles the boundary overlaps, weighted by overlap area — not a centroid lookup. At 60 m the parcel spans roughly 3.8 tiles, so a nearest-tile reading would discard most of the site.

Why duration leads. Across this AOI the daily-peak snapshot varies by only 0.90 °C, while hours-above-threshold varies by 15.2 h. At parcel scale the duration layers, not the snapshot, carry the signal.

FINDINGS AND RECOMMENDED ACTIONS

Each row is a published threshold, the measured value against it, and the program the action comes from. No composite index is used.

Trigger	Measured	Threshold	Verdict
Hours above 27 °C over 168 h	35.6 h (21% of window)	27 °C (NOAA Caution)	ELEVATED
Longest consecutive run above threshold	6.5 h	12 h (overnight recovery)	RECOVERS OVERNIGHT
Impervious surface share (site vicinity)	26.9%	60%	WITHIN
Tree canopy share (site vicinity)	1.6%	15%	BELOW
Ground-level vegetation at frontage	6.5% of field of view	10% (pedestrian shade)	BELOW
Heat index at the hot hour (13:00)	27.3 °C	32 °C (Extreme Caution)	CAUTION BAND
Construction-phase worker exposure at peak hour	27.3 °C at 13:00	32.2 °C (OSHA high-heat)	NOT TRIGGERED
Peak wet-bulb temperature	17.1 °C	26 °C (evaporative-cooling limit)	BELOW

Hours above 27 °C over 168 h — ELEVATED

Cooling load is duration-driven, not peak-driven; model annual hours, and specify envelope for sustained rather than instantaneous load.

Program: ASHRAE 90.1 / ASHRAE 55 — Thermal Environmental Conditions

Longest consecutive run above threshold — RECOVERS OVERNIGHT

Exposed thermal mass remains a viable passive strategy on this site.

Program: EPA Heat Island Reduction — Cool Roofs

Impervious surface share (site vicinity) — WITHIN

No cool-surface retrofit indicated by surface share alone.

Program: EPA Heat Island Reduction

Tree canopy share (site vicinity) — BELOW

Include a planting plan in the landscape package; quantify the canopy benefit before entitlement hearings.

Program: USDA Forest Service — i-Tree

Ground-level vegetation at frontage — BELOW

Frontage reads as unshaded at eye level — street trees or a shade structure are needed for any ground-floor active use.

Program: NACTO Urban Street Design Guide

Heat index at the hot hour (13:00) — CAUTION BAND

Peak-hour heat index sits in the NOAA Caution band — tolerable, but shaded outdoor amenity is still indicated for sustained use.

Program: ASHRAE 55 — Thermal Environmental Conditions

Construction-phase worker exposure at peak hour — NOT TRIGGERED

Not triggered on this study day. The site does spend 36 h above 27 °C air temperature per 168 h (Step 5), so re-check against a summer design day before ruling it out for the build window.

Program: OSHA Heat Illness Prevention

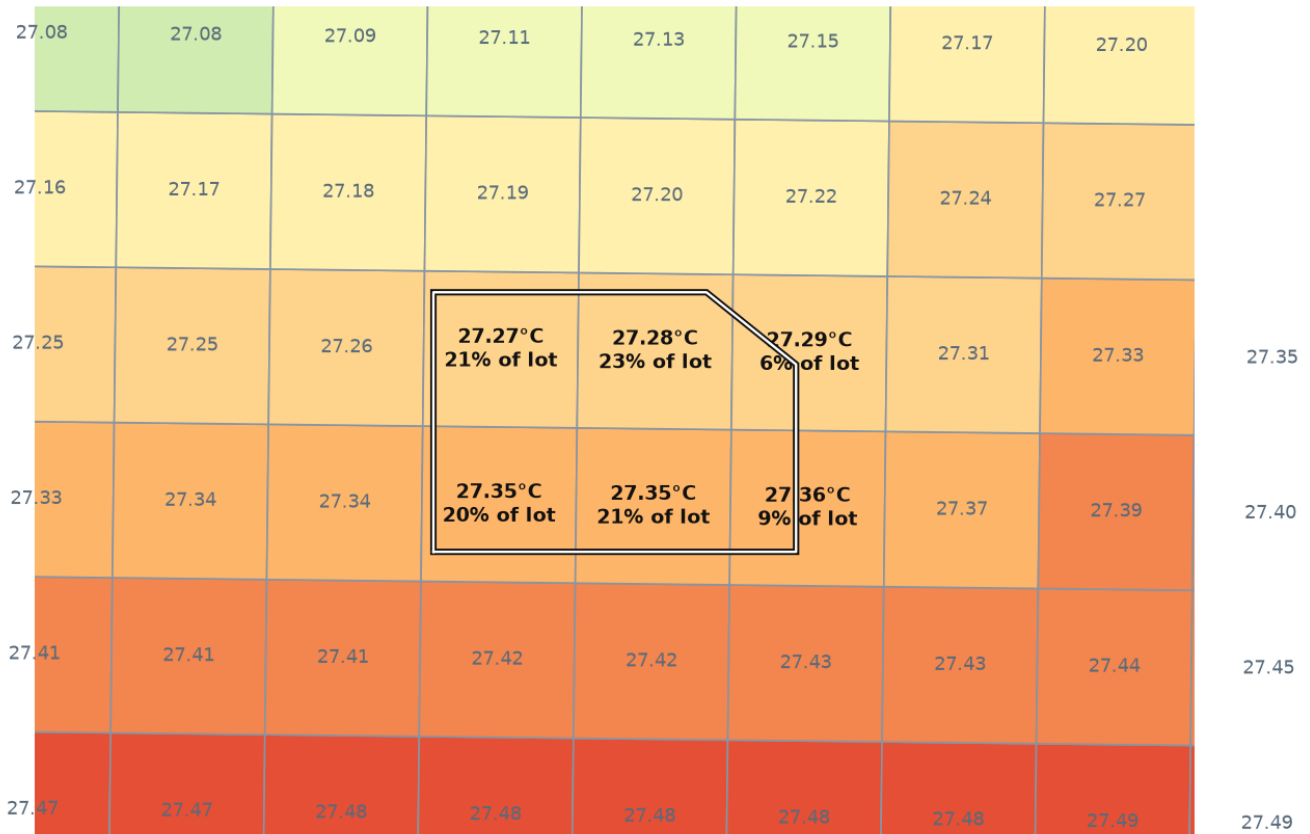
Peak wet-bulb temperature — BELOW

Evaporative and hybrid cooling strategies remain viable on this site.

Program: ASHRAE Handbook — HVAC Applications

SITE CONTEXT AND HEAT LAYER

Area-weighted clip — parcel peak 27.31 °C from 6 overlapping tiles at 60 m

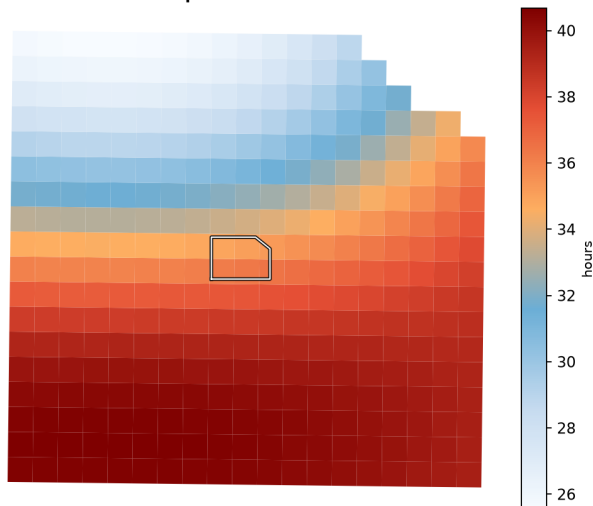


Area-weighted clip: every tile the boundary overlaps, with its share of the lot.

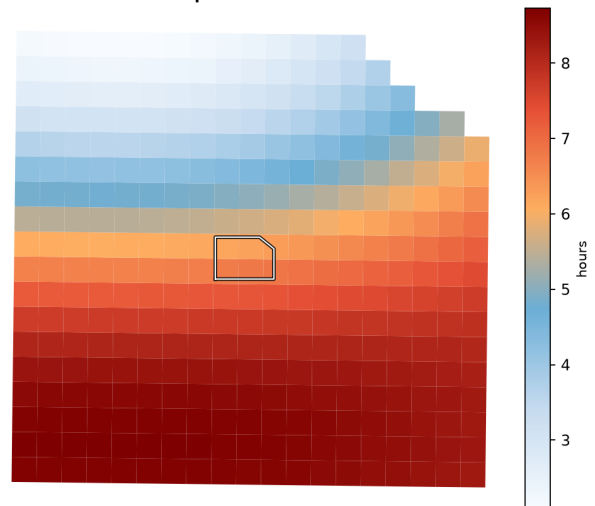
EXPOSURE DURATION

Diridon Gateway Site — exposure duration

Hours above 27 °C
2024-07-12 .. 2024-07-18 (168 h window)
parcel: 35.6 h

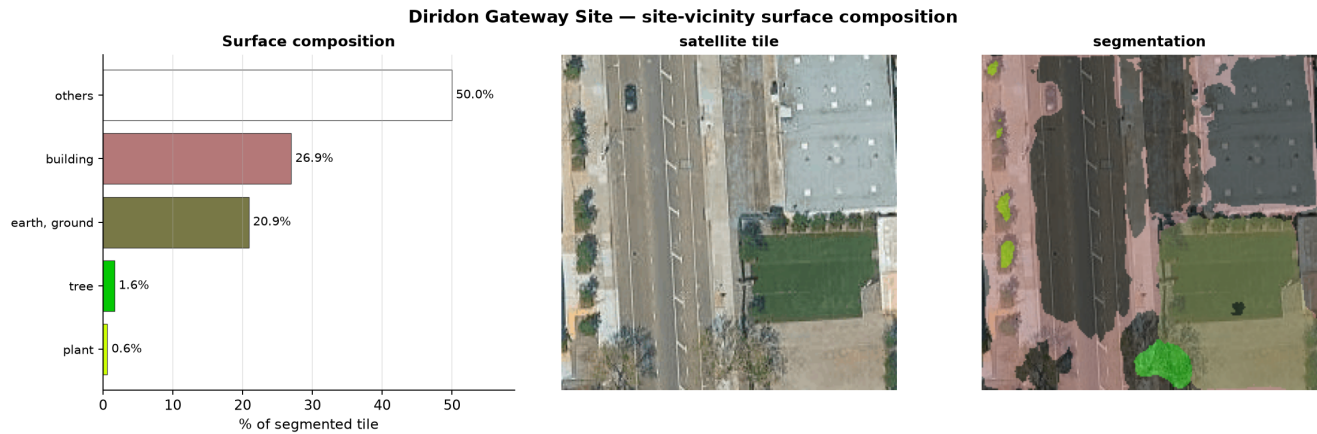


Longest consecutive run above 27 °C
2024-07-12 .. 2024-07-18
parcel: 6.5 h



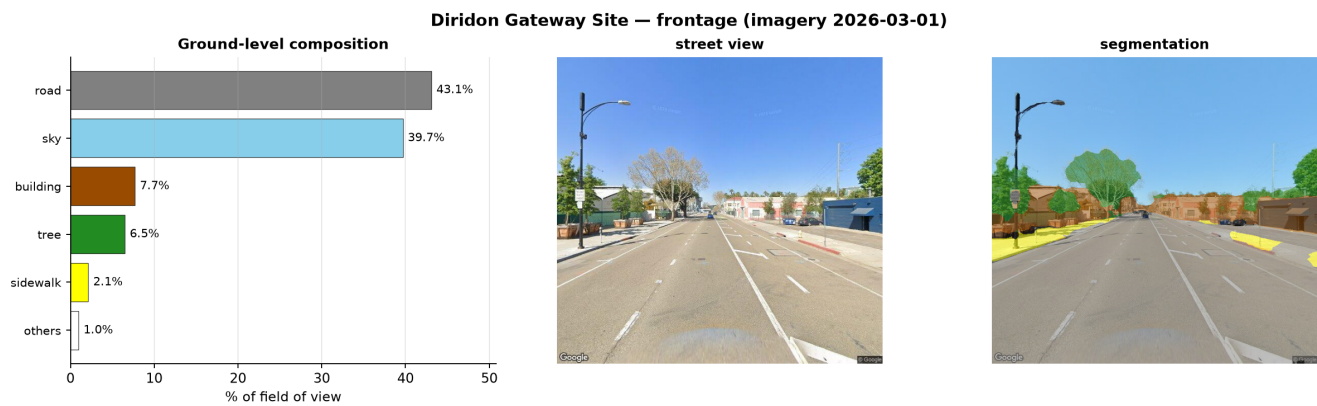
Hours above 27 °C and longest consecutive run, 2024-07-12 to 2024-07-18.

SURFACE COMPOSITION



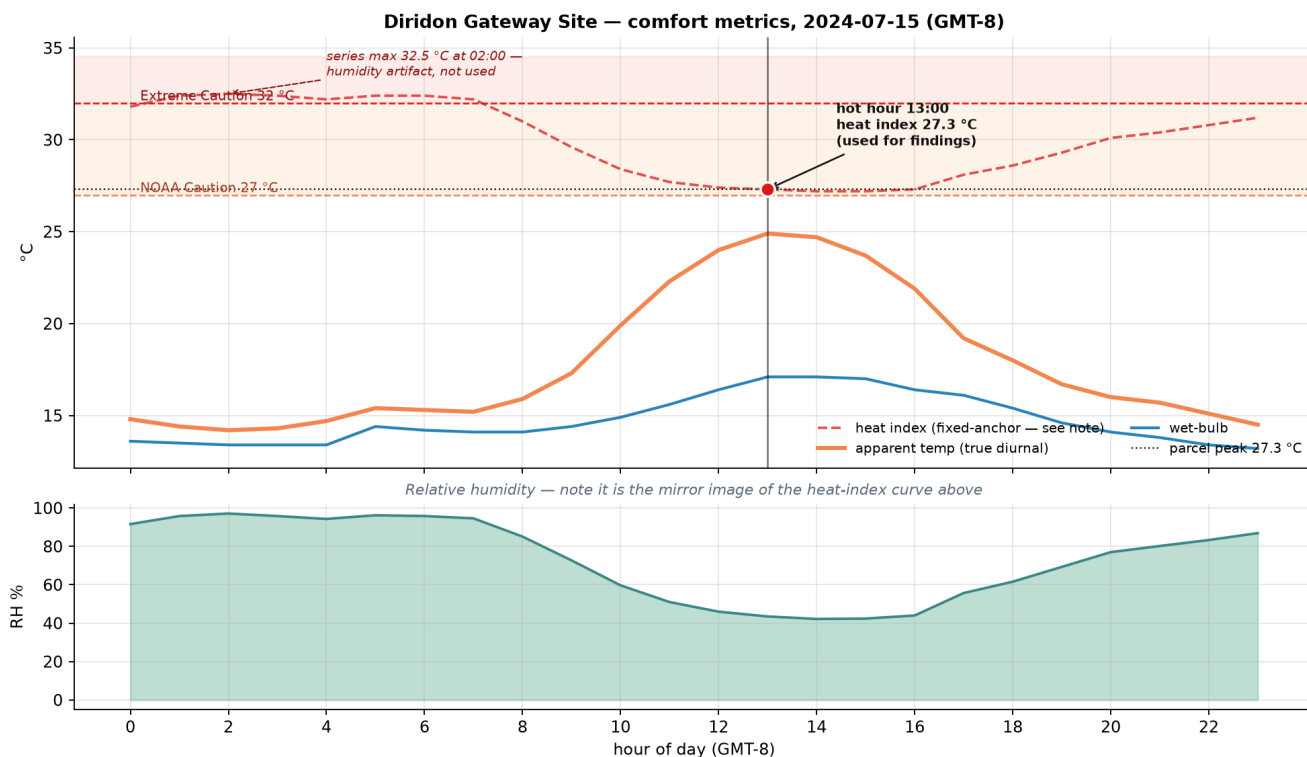
Satellite land-cover segmentation at the site vicinity.

FRONTAGE



Street-view segmentation — the vertical environment at ground level.

COMFORT METRICS



Apparent temperature follows the true diurnal cycle. The heat-index curve (dashed) is derived from a fixed temperature anchor and varies only with humidity, so it inverts — only its value at the hot hour is used.

SOURCES AND LIMITS

Thresholds. NOAA heat-index Caution (27 °C) and Extreme Caution (32 °C); OSHA high-heat trigger (32.2 °C); EPA Heat Island Reduction cool-roof / cool-pavement guidance; USDA Forest Service i-Tree canopy targets; ASHRAE 90.1 and ASHRAE 55.

Data. FortyGuard tOS Enterprise API. Heatmap 2024-07-15 at 60 m; exposure duration 2024-07-12 to 2024-07-18; satellite, street-view, and environmental parameters at the parcel centroid.

Limits. Satellite and street-view segmentation are anchored to the parcel centroid and describe the site vicinity, not the parcel polygon exactly. The heatmap resolves to 60 m, coarser than the parcel, so within-site variation is not resolved. Comfort metrics derive from a single study day and should be re-run against a summer design day before they inform final mechanical sizing.

Heat index. The environmental-parameters endpoint applies one temperature anchor across all 24 hours and varies only humidity, so the heat-index series tracks humidity and peaks overnight — an artifact, not a forecast. This report uses heat index only at the hot hour (13:00, 27.3 °C), where the anchor reflects real conditions. All duration figures come from the measured heatmap exceedance layer, not from this series.